

ACETIC ACID

ICSC: 0363 (May 2010)

Glacial acetic acid
 Ethanoic acid
 Ethylic acid
 Methanecarboxylic acid

CAS #: 64-19-7

UN #: 2789

EC Number: 200-580-7

	ACUTE HAZARDS	PREVENTION	FIRE FIGHTING
FIRE & EXPLOSION	Flammable. Above 39°C explosive vapour/air mixtures may be formed. Risk of fire and explosion on contact with strong oxidants.	NO open flames, NO sparks and NO smoking. Above 39°C use a closed system, ventilation and explosion-proof electrical equipment.	Use powder, alcohol-resistant foam, water spray, carbon dioxide. In case of fire: keep drums, etc., cool by spraying with water.

AVOID ALL CONTACT!			
	SYMPTOMS	PREVENTION	FIRST AID
Inhalation	Sore throat. Cough. Burning sensation. Headache. Dizziness. Shortness of breath. Laboured breathing.	Use ventilation, local exhaust or breathing protection.	Fresh air, rest. Half-upright position. Refer immediately for medical attention.
Skin	Pain. Redness. Skin burns. Blisters.	Protective gloves. Protective clothing.	Remove contaminated clothes. Rinse and then wash skin with water and soap. Rinse skin with plenty of water or shower for at least 15 minutes. Refer immediately for medical attention.
Eyes	Redness. Pain. Severe burns. Loss of vision.	Wear face shield or eye protection in combination with breathing protection.	Rinse with plenty of water (remove contact lenses if easily possible). Refer immediately for medical attention.
Ingestion	Sore throat. Burning sensation. Abdominal pain. Vomiting. Shock or collapse.	Do not eat, drink, or smoke during work.	Rinse mouth. Do NOT induce vomiting. If within a few minutes after ingestion, one small glass of water may be given to drink. Refer immediately for medical attention.

SPILLAGE DISPOSAL	CLASSIFICATION & LABELLING
Remove all ignition sources. Personal protection: chemical protection suit including self-contained breathing apparatus. Do NOT let this chemical enter the environment. Collect leaking liquid in sealable containers. Cautiously neutralize spilled liquid with sodium carbonate only under the responsibility of an expert.	According to UN GHS Criteria  DANGER Flammable liquid and vapour Harmful in contact with skin or if inhaled May be harmful if swallowed Causes severe skin burns and eye damage May cause respiratory irritation Causes damage to the respiratory tract through prolonged or repeated exposure if inhaled Harmful to aquatic life Transportation UN Classification UN Hazard Class: 8; UN Subsidiary Risks: 3; UN Pack Group: II
STORAGE	
Fireproof. Separated from food and feedstuffs, strong oxidants, strong acids and strong bases. Store only in original container. Well closed. Keep in a well-ventilated room. Store in an area without drain or sewer access.	
PACKAGING	
Do not transport with food and feedstuffs.	



Prepared by an international group of experts on behalf of ILO and WHO, with the financial assistance of the European Commission.
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ACETIC ACID**ICSC: 0363****PHYSICAL & CHEMICAL INFORMATION****Physical State; Appearance**

COLOURLESS LIQUID WITH PUNGENT ODOUR.

Physical dangers

No data.

Chemical dangers

The substance is a weak acid. Reacts violently with strong oxidants. This generates fire and explosion hazard. Reacts violently with strong bases, strong acids and many other compounds. Attacks some forms of plastic, rubber and coatings.

Formula: $C_2H_4O_2$ / CH_3COOH

Molecular mass: 60.1

Boiling point: 118°C

Melting point: 16.7°C

Relative density (water = 1): 1.05

Solubility in water: miscible

Vapour pressure, kPa at 20°C: 1.5

Relative vapour density (air = 1): 2.1

Relative density of the vapour/air-mixture at 20°C (air = 1): 1.02

Flash point: 39°C c.c.

Auto-ignition temperature: 485°C

Explosive limits, vol% in air: 6.0-17

Octanol/water partition coefficient as log Pow: -0.17

EXPOSURE & HEALTH EFFECTS**Routes of exposure**

Serious local effects by all routes of exposure.

Effects of short-term exposure

The substance is corrosive to the eyes, skin and respiratory tract. Corrosive on ingestion. Inhalation may cause lung oedema, but only after initial corrosive effects on eyes and/or airways have become manifest.

Inhalation risk

A harmful contamination of the air can be reached rather quickly on evaporation of this substance at 20°C.

Effects of long-term or repeated exposure

Repeated or prolonged contact with skin may cause dermatitis. Lungs may be affected by repeated or prolonged exposure to an aerosol of this substance. Risk of tooth erosion upon repeated or prolonged exposure to an aerosol of this substance.

OCCUPATIONAL EXPOSURE LIMITSEU-OEL: 25 mg/m³, 10 ppm as TWA; 50 mg/m³, 20 ppm as STEL.MAK: 25 mg/m³, 10 ppm; peak limitation category: I(2); pregnancy risk group: C.

TLV: 10 ppm as TWA; 15 ppm as STEL

ENVIRONMENT

The substance is harmful to aquatic organisms.

NOTES

UN number 2789 is for acetic acid, glacial or acetic acid solution, more than 80% acid by mass.

Other UN numbers: UN 2790 acetic acid solution (10-80% acetic acid); UN hazard class 8, packing group II-III.

ADDITIONAL INFORMATION**EC Classification**

Symbol: C; R: 10-35; S: (1/2)-23-26-45; Note: B

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